Skin Disease Classifier

Student Group Project Report

Applied Image Processing

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Prepared by: Group 19

Anurag Thakur 2023IMG-008

Baviskar Pratik Subhash 2023IMG-014

Kshitij Dhamanikar 2023IMG-029

Kushagra Agarwal 2023IMG-030

Vibhor Kumar 2023IMG-054

Abstract

This report presents a complete machine learning workflow for automated skin disease image classification. The final system uses a Vision Transformer (ViT)-based weighted ensemble for 8-class prediction. Compared to a single promoted ViT baseline, the final ensemble improves overall accuracy and macro-level metrics substantially. The deployment-ready artifacts were integrated into a Flask web application runtime and validated through class-wise metrics, confusion analysis, and leakage-aware evaluation checks.

Final ensemble metrics on the main test split: Accuracy=0.8533, Macro Precision=0.8610, Macro Recall=0.8533.

Keywords

Skin disease classification, Vision Transformer, Ensemble learning, Medical image analysis, Deep learning

1. Introduction

Skin disorders are common and often require timely assessment. Automated classification tools can support screening, education, and decision support when designed with clear limitations. This project focuses on building a practical, reproducible image classification pipeline under CPU-only constraints while maintaining strong model quality.

1.1 Problem Statement

Given an input skin image, classify the disease category among 8 predefined classes: acne, bcc, bkl, eczema, ak, melanoma, psoriasis, and tinea.

1.2 Objectives

- Build CPU-friendly ViT training and evaluation workflows.
- Improve model quality through controlled ensemble optimization.
- Integrate final model configuration into application runtime.
- Produce clear metrics, plots, and report artifacts for academic submission.
- Validate performance with leakage-aware checks for reliability.

1.3 Scope and Limitations

The system is intended for educational and research use and is not a replacement for clinical diagnosis. Model performance is dataset-dependent and can degrade on unseen data domains.

2. Dataset and Experimental Setup

Experiments were run on curated split folders with balanced class representation in the test split. The final evaluated split consists of 184 total images with 23 images per class.

Item	Value
Primary evaluation split	datasets/hf_hybrid_150/splits/test
Number of classes	8
Total test samples	184
Samples per class	23
Runtime environment	CPU-only training/evaluation
Model family	Vision Transformer (ViT)

2.1 Leakage-Aware Evaluation Note

Because some dataset versions may share source images, leakage-safe subset checks were used to validate whether gains persist after excluding overlapping train/validation references from expanded dataset variants.

2.2 Experimental Protocol

- Train multiple ViT checkpoints with varied augmentation and regularization profiles.
- Evaluate checkpoints on common split for comparability.
- Search ensemble weights for improved accuracy and stability.
- Promote best-performing ensemble into runtime config.

3. Model Architecture and Pipeline

The final predictor is a weighted softmax ensemble over four ViT checkpoints. Each member contributes probabilities according to fixed weights, and the class with maximum weighted probability is selected.

Ensemble Member	Model Path	Weight
strict80_tiny160	artifacts/ vit_strict80_tiny160/ best_model.pt	0.50
220_tiny224_lowaug	artifacts/ vit_220_tiny224_lowaug/ best_model.pt	0.40
150_tiny224_lowaug	artifacts/ vit_150_tiny224_lowaug/ best_model.pt	0.05
150_tiny160	artifacts/vit_150_tiny160/ best_model.pt	0.05

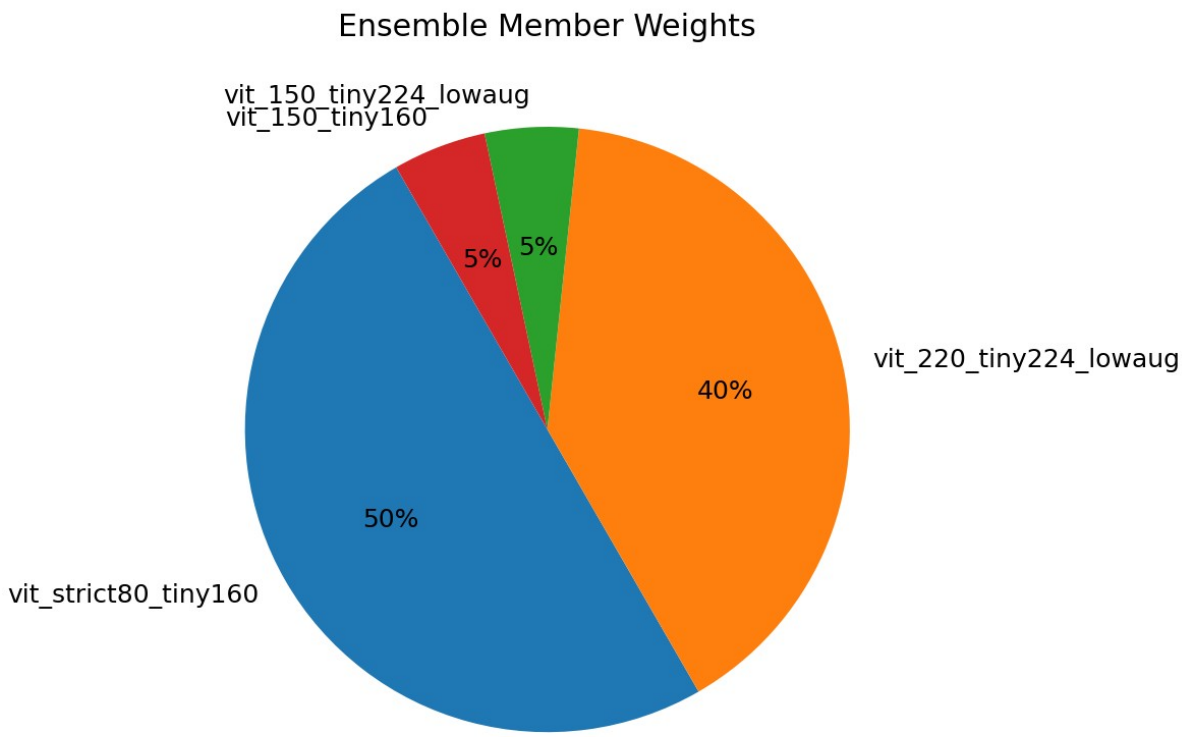

Figure 1: Final ensemble member weight distribution.

3.1 Deployment Path

- Runtime config: website/models/vit_ensemble.json
- Metrics export: website/models/vit_ensemble_metrics.json
- Confusion export: website/models/vit_ensemble_confusion.csv
- Serving interface: Flask application inference route

4. Final Quantitative Results

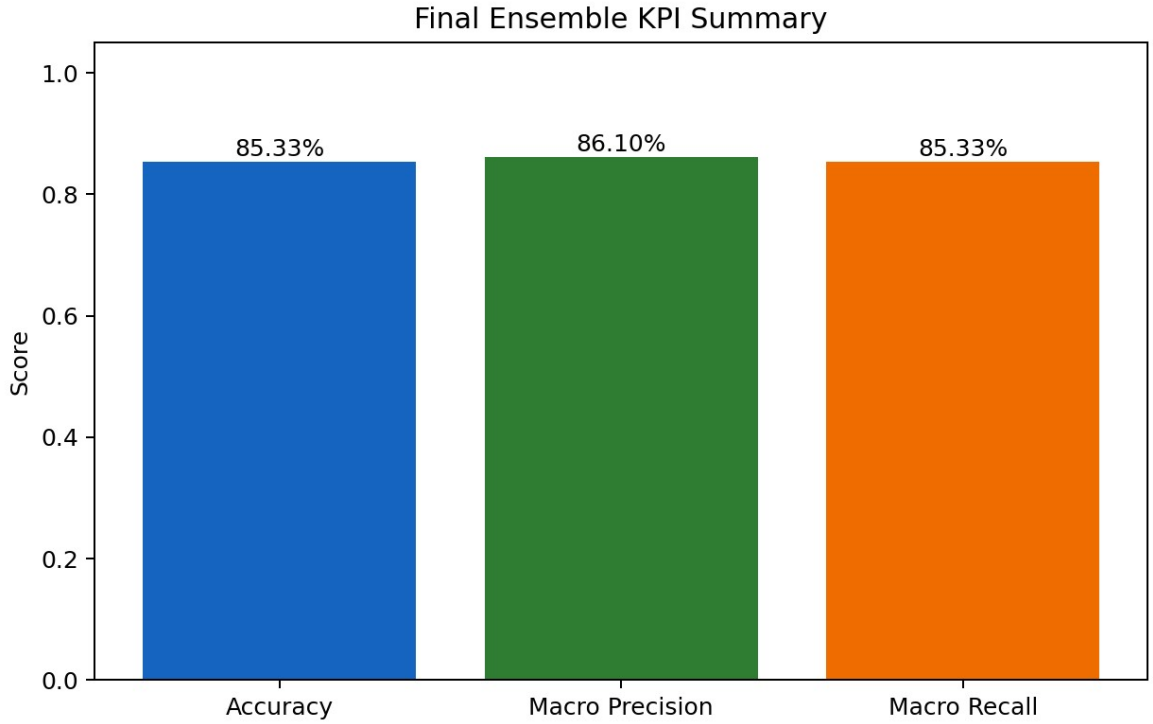

Figure 2: KPI summary of final ensemble.

Metric	Final Ensemble	Single ViT Baseline	Absolute Gain
Accuracy	0.8533	0.7120	+0.1413
Macro Precision	0.8610	0.7036	+0.1575
Macro Recall	0.8533	0.7120	+0.1413

The ensemble gives a clear and consistent improvement over the single baseline model across all primary aggregate metrics.

4.1 Class-wise Metrics

Class	Precision	Recall	Support
acne	0.7931	1.0000	23
bcc	0.8800	0.9565	23
bkl	0.9091	0.8696	23
eczema	0.8750	0.6087	23
ak	0.9130	0.9130	23
melanoma	0.9091	0.8696	23
psoriasis	0.7143	0.8696	23
tinea	0.8947	0.7391	23

5. Diagnostic Analysis

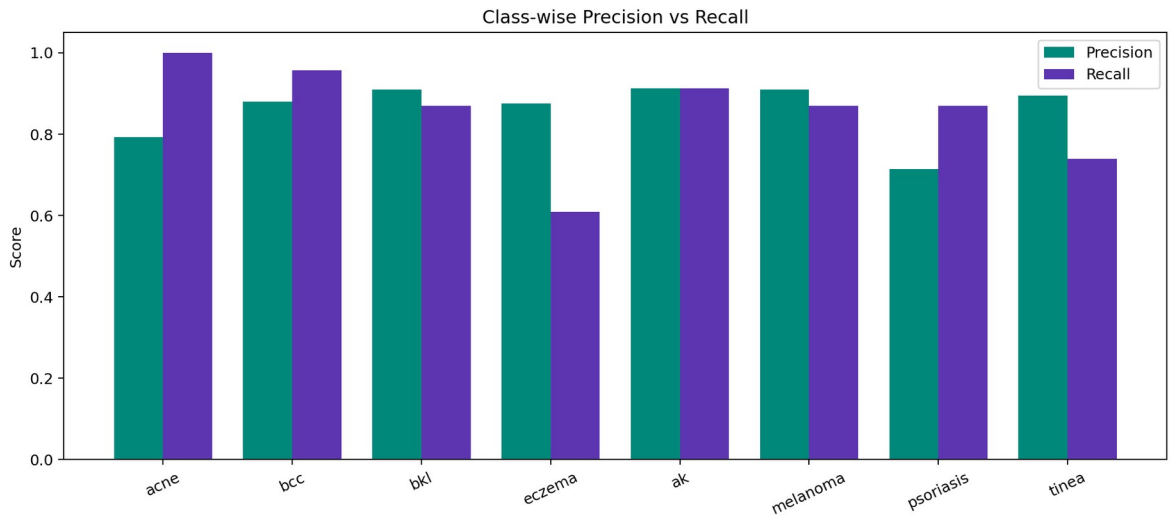

Figure 3: Precision and recall by class.

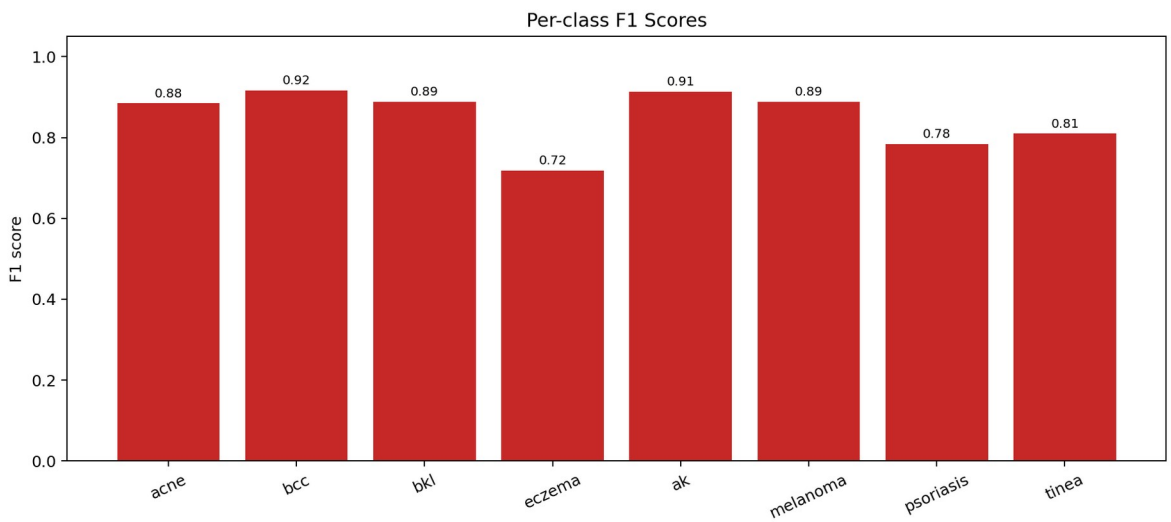

Figure 4: Per-class F1 score.

5.1 Key Findings

- Acne and BCC show strong recall and stable precision.
- AK and Melanoma maintain balanced performance.
- Eczema remains the most difficult class in recall.
- Tinea has high precision but lower recall, suggesting conservative prediction behavior.

6. Confusion and Prediction Distribution

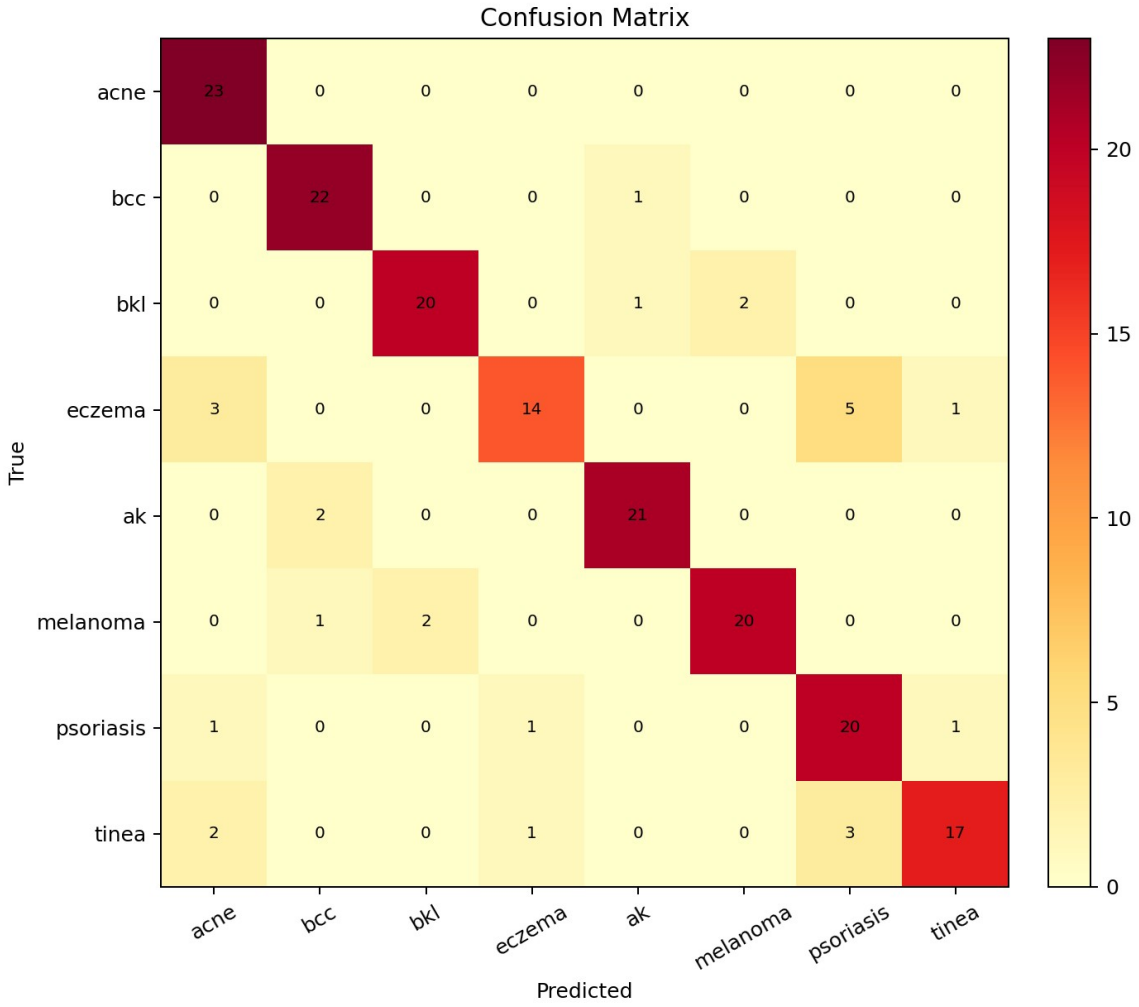

Figure 5: Confusion matrix heatmap.

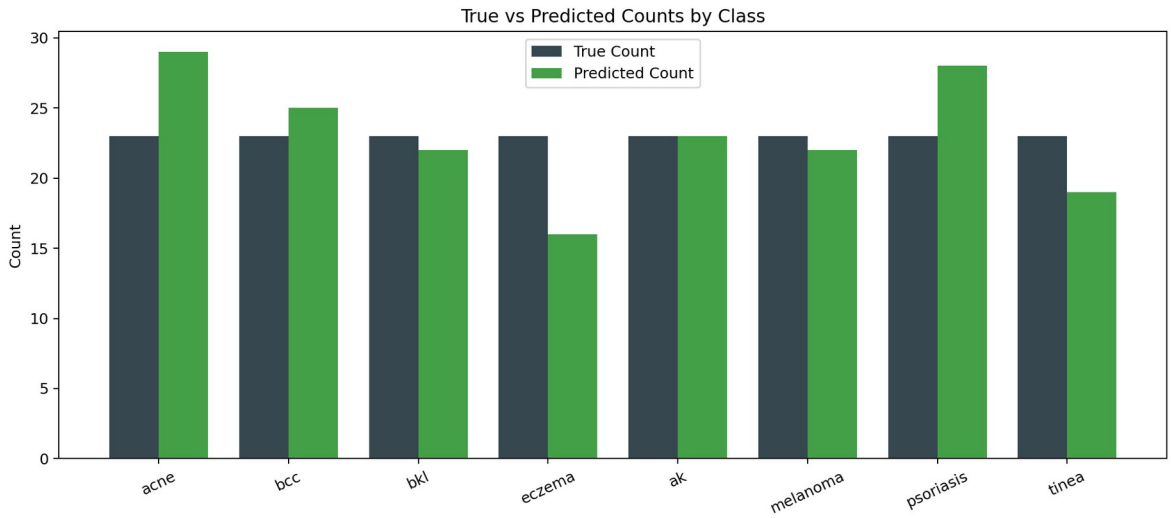

Figure 6: True vs predicted class counts.

6.1 Confusion Patterns

- Eczema has notable confusion with Psoriasis and Acne.
- AK occasionally overlaps with BCC.
- Tinea has occasional confusion with Psoriasis and Acne.
- Acne reached perfect recall in the final run.

6.2 Raw Confusion Table

true/ pred	acne	bcc	bkl	eczema	ak	melanoma	psoriasis	tinea
acne	23	0	0	0	0	0	0	0
bcc	0	22	0	0	1	0	0	0
bkl	0	0	20	0	1	2	0	0
eczema	3	0	0	14	0	0	5	1
ak	0	2	0	0	21	0	0	0
melanoma	0	1	2	0	0	20	0	0
psoriasis	1	0	0	1	0	0	20	1
tinea	2	0	0	1	0	0	3	17

7. Risks, Ethics, and Limitations

- Performance remains below 90% target under CPU constraints.
- Visually similar classes can still produce boundary errors.
- Dataset overlap management remains critical for fair benchmarking.
- Uncertainty calibration and abstention are not yet implemented.

7.1 Ethical Note

This model is intended for educational and research use only. It is not a medical device and should not be used as a standalone diagnostic tool.

8. Conclusion and Future Work

The project successfully delivered an integrated ViT ensemble pipeline with strong improvements over baseline. The generated artifacts include deployable configuration files, quantitative reports, confusion diagnostics, and reproducible scripts.

- Train higher-capacity ViT backbones on strict non-overlap datasets.
- Introduce confidence calibration for safer prediction handling.
- Perform external benchmark validation for robustness estimation.
- Add monitoring hooks for model drift in deployment.

Appendix A: Artifact References

Artifact	Purpose
website/models/vit_ensemble.json	Final runtime ensemble configuration
website/models/vit_ensemble_metrics.json	Primary final metrics
website/models/vit_ensemble_confusion.csv	Confusion matrix values
docs/report_assets/*.png	Figures used in report and presentation
scripts/generate_project_report_docx.py	DOCX report generator